

PUBLIC ARCHIVE / VERSION 3.1 / CORRECTED INTERPRETATION

State-Aware Computing

Exact local reuse for repeated system intervention

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Abstract

State-Space Programming Methodology (SSPM) is a restricted state-transition runtime for repeated interventions on a shared system. It keeps a validated baseline state resident, edits a local target, propagates the complete declared dependency closure, and reuses unaffected values. If locality disappears, the runtime falls back to full replay. Under sufficient-state and complete-dependency conditions, compact execution is consistent with from-scratch execution. Across traffic, queue, thermal, branching, and workflow families, sparse closures often reduced local CPU decision cost; dense, cold, and materialization-heavy regimes did not. A fair cross-project control later reproduced the same adaptive transition work. The result is a bounded execution and state-organization mechanism, not representation-level causality, universal linearization, or universal speed.

1. Transition semantics

Let x_t be sufficient declared state, u_t the exogenous input, and F the ordered transition:

$$x_{t+1} = F(x_t, u_t).$$

A from-scratch rollout produces a baseline trajectory x^0 . An intervention changes a subset of state, controls, or parameters. SSPM computes the transitive affected closure C_t induced by declared reads, writes, ordering, and coupling. At each step, values in C_t are recomputed and values outside it are reused from x^0 . A demanded output is reconstructed from both regions.

Conditional exactness. Compact execution equals full replay if state is sufficient; all dependencies are declared; the closure contains every value reachable from the edit; reused values share transition semantics, parameters, and exogenous inputs; and output reconstruction is complete. An omitted dependency or hidden side effect breaks the premise. Full replay is therefore a correctness-preserving runtime path, not merely a performance baseline.

A complete proof models the rollout as a time-expanded dependency graph and uses topological induction to establish unaffected-node invariance and recomputed-node equality. See Formal Semantics and Conditional Exactness of State-Aware Transition Reuse.

STATE-AWARE EXECUTION MODEL

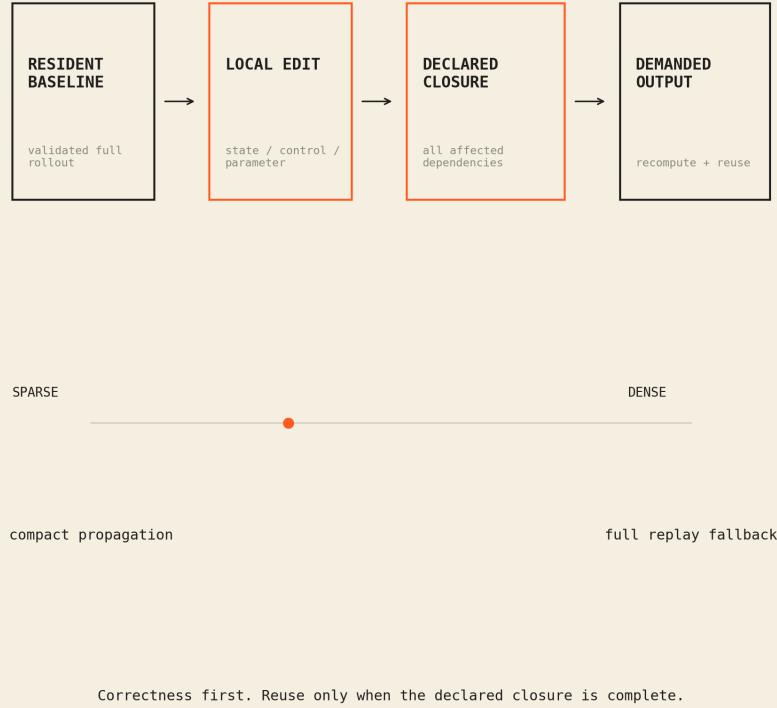


Figure 1: State-aware execution model

2. Mechanism and novelty boundary

The mechanism is designed for repeated “what changes if this changes?” queries. One baseline is computed and retained. Each candidate intervention pays for closure discovery, compact propagation, and required materialization rather than re-building and replaying the entire system. Its useful region is:

$$T_{closure} + T_{compact} + T_{materialize} < T_{full}.$$

The work sits near incremental computation, self-adjusting computation, dynamic dependence graphs, and differential data systems [1-3]. SSPM’s narrower focus is transition-oriented numerical systems with explicit state semantics, local intervention, demanded-output reconstruction, and a measured full-replay fallback. The current source frontend covers four supported grammar families across 60 parameterized programs. It is not a general compiler and does not claim 60 independent applications.

This organization does not make selective propagation unique to SSPM. An equally capable conventional runtime can track the same dependencies and perform the same transition work. The research distinction must therefore be tested in compact branch storage, validation, materialization, governance, and the integration of exact fallback rather than inferred from representation alone.

3. Evidence from V4-V13

The research cycles first established exact resident incremental execution, then tested transfer, generated execution, closure boundaries, stronger external backends, multiple applications, and external workflow traces. All performance claims were gated on differential consistency and checksummed frozen evidence. After correcting affine classification so nonlinear products remain residual, the frozen V13 corpus still accepts 40 and rejects 20 parameterized programs with zero maximum error among accepted cases.

V12 provides the clearest paired boundary. Compact execution won 14 of 16 sparse application rows; the median advantage among those wins was 4.36x. Full replay won all seven dense rows, with a 3.44x median advantage. Traffic, dynamic queue, thermal, and branching families therefore support conditional transfer without supporting a universal speed claim.

V13 evaluates external Montage workflow traces. The resident decision path won 23 of 24 rows, with a 5.58x median advantage among wins. At the broader measured workflow boundary, 21 of 24 rows won, with a 4.43x median advantage among wins. The distinction matters: the first is resident-only and the second includes the measured end-to-end workflow costs.

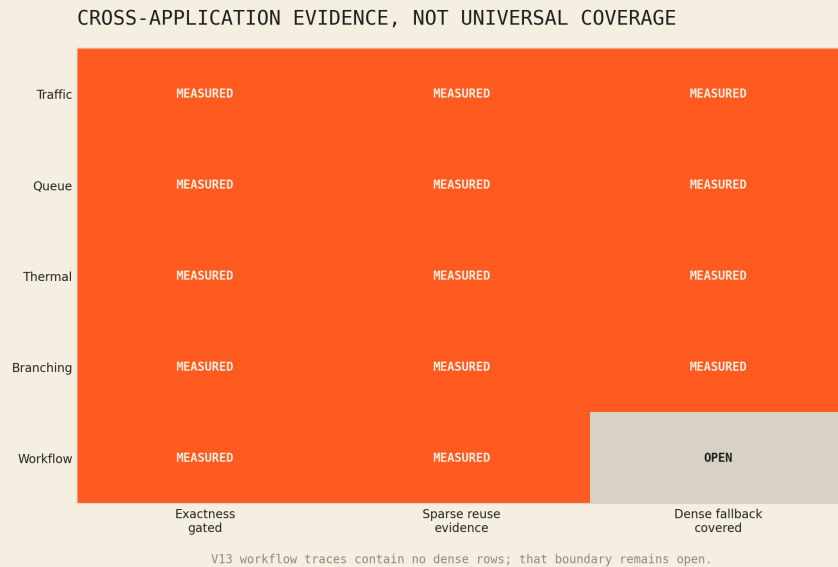


Figure 2: Cross-application evidence

4. Negative results and disposition

The broad V11 median was 1.82x slower than the strongest compiled-batch baseline even though five qualifying sparse rows produced a 2.32x median advantage. Cold JIT cost did not amortize through 10,000 tested decisions. Dense V12 rows favored full replay, and V13 contains no dense rows. Differential Dataflow, isolated reproduction, CUDA/Triton, and multi-GPU gates remain incomplete or deferred.

The V13 classifier also requires a strict label: it consumes realized closure after compact execution. It is a post-hoc locality classifier, not an operational pre-execution selector.

Cross-project E2A evidence provides a second correction boundary. Case 14 preserved semantics and compact branches but failed efficiency at 1.2122x executed work. Case 15 gave the conventional control equivalent dependency tracking, value stabilization, and adaptive propagation; executed work became equal at 1.0000x. Its lower inspection and branch-storage ratios remain bounded state-organization results, not proof of intrinsic acceleration. The public E2A archive preserves the correction and source boundaries.

The evidence supports **state-aware computing** as a bounded method: share a resident baseline, recompute the complete affected closure, reconstruct only what is demanded, and use full replay when locality or economics disappear. It does not support arbitrary-system linearization, hidden-state reuse, or a claim that incremental execution is always the fastest path. The frozen SSPM ratios remain measurements of their declared workloads, not a causal comparison between state-space and conventional representation.

References

- [1] U. A. Acar et al., "An Experimental Analysis of Self-Adjusting Computation," *ACM TOPLAS*, 32(1), 2009. <https://doi.org/10.1145/1596527.1596530>
- [2] U. A. Acar, G. E. Blelloch, and R. Harper, "Adaptive Functional Programming," *ACM TOPLAS*, 28(6), 2006. <https://doi.org/10.1145/1186632.1186634>
- [3] F. McSherry et al., "Differential Dataflow," *CIDR*, 2013. https://www.cidrdb.org/cidr2013/Papers/CIDR13_Paper111.pdf

Evidence boundary. Aggregate local CPU results under declared workloads and timing boundaries. No raw logs, private paths, host identifiers, live systems, or accelerator-performance claims are included here. Executable evidence: public SSPM workbench.